4 A copper ball of mass m is attached to the top of a spring of spring constant k. They are placed on a flat, horizontal table, as shown in Fig. 4.1. The copper ball is in equilibrium and the spring is compressed.

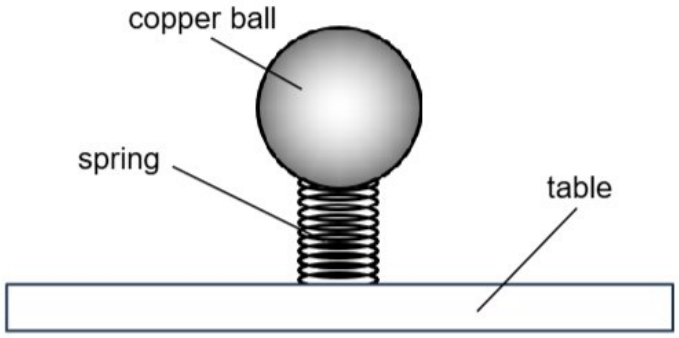


Fig. 4.1

The copper ball is pushed downwards such that its displacement is x_0 below its equilibrium position and then released.

(a) State and explain, by considering the forces acting on the copper ball, the direction of the resultant force acting on the copper ball immediately after its release.

.....

.....

..... [2]

(b) Show that the acceleration a of the copper ball is given by the expression:

$$a = -\frac{k}{m}x$$

where x is the vertical displacement of the copper ball from its equilibrium position.

[2]

(c) Using the expression in **(b)**, explain why the copper ball undergoes simple harmonic oscillations.

.....

.....

..... [2]

(d) A student conducted an experiment using the setup in Fig 4.1 and obtained the following measurements:

$$m = (1.200 \pm 0.005)\ \text{kg}$$
$$k = (13.6 \pm 0.2)\ \text{N m}^{-1}$$

The angular velocity ω of the oscillations is given by the expression

$$\omega = \sqrt{\frac{k}{m}}$$

(i) Calculate the fractional uncertainty for ω.

fractional uncertainty = ... [2]

(ii) Hence, calculate ω., together with its absolute uncertainty.

ω = ...± ..... rad s^{-1} [2]

(iii) Determine the total energy of the oscillation when x_0 is measured to be 0.20 m.

total energy = ... J [2]

[Total:12]